MILITARY SPECIFICATION

DISPERSER, RIOT CONTROL AGENT, HELICOPTER- OR VEHICLE-MOUNTED, M4

1. SCOPE

1.1 This specification covers the M4 helicopter- or vehicle-mounted riot control agent disperser.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-B-571	-	Bolts, Nuts, Studs, and Tap-Rivets (and Material for Same).
NN-P-515	-	Plywood, Container Grade.
QQ-S-636	-	Steel, Carbon (Low-Carbon), Sheets and Strips.
QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon.
QQ-S-781	-	Steel Strapping, Flat.
LLL-H-35	-	Hardboard, Fibrous-Felted (Fiberboard).
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Set-Up, Paperboard.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-T-60	-	Tape; Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing.
PPP-T-97	-	Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-C-104	-	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	-	Preservation, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-F-26862	-	Fiberboard, Solid, Dunnage, Multipurpose Cushioning and Blocking Applications.
MIL-T-50036	-	Talc, Technical, T1.
MIL-G-51025	-	Gun, Portable Riot Control Agent Disperser, M9.

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

CHEMICAL CORPS

DL 116-6-75	-	Disperser, Riot Control Agent, Helicopter- or Vehicle-Mounted, M4. (All drawings referenced in this specification are listed on the above drawing.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL 116-6-75 or on the subsidiary drawings shall conform to the specifications thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the disperser shall conform to the specifications and drawings listed on Drawing DL 116-6-75 and subsidiary drawings listed thereon.

3.2 Assembly. The disperser shall be assembled as shown on Drawing DL 116-6-75.

3.3 Leakage.

3.3.1 Tank and frame assembly. The tank and frame assembly, Drawing D 116-6-88, with tank adapter assembly removed shall not leak when tested as specified in 4.7.1.1.

3.3.2 Assembled disperser, air and agent. The assembled disperser, Drawing DL 116-6-75, shall not leak when tested as specified in 4.7.1.2.

3.4 Safety head. The safety head rupture diaphragm shall rupture between 80 and 100 pounds per square inch gage (psig) when tested as specified in 4.7.2.

3.5 Preproduction sample. Prior to the start of regular production, a preproduction sample disperser shall be produced using the same methods, materials, and equipment as will be used during regular production.

3.6 Functioning. Each assembled disperser, Drawing DL 116-6-75, shall discharge a minimum of 29 pounds of simulant agent in a maximum period of 45 seconds when tested as specified in 4.7.3.

3.7 Workmanship. The assembled disperser shall be free from damage, such as burrs, kinks, dents, or other defects that would adversely affect its functioning or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves

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the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government's responsibility. The Government will witness the test specified in 4.7.3.

4.2 Special provisions.

4.2.1 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedure that will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification shall govern.

4.2.2 Supplemental evidence. The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.3 Preproduction sample inspection.

4.3.1 Examination. The sample disperser shall be inspected for all the requirements of the applicable drawings and this specification.

4.3.2 Test. The sample disperser shall also be tested in accordance with 4.7.3.

4.3.3 Acceptance/rejection criteria. The sample disperser shall meet the examination and test specified in 4.3.1 and 4.3.2 respectively to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Lotting. A lot shall consist of the safety heads produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions and offered for acceptance at one time.

4.5 Sampling.

4.5.1 Nondestructive examination. Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.5.2 Destructive test. Four percent of the safety head rupture discs shall be taken at random from each regular production lot.

4.6 Examination. Each disperser shall be examined for all the requirements of this specification. Sample components and the preparation for delivery (level A) shall be examined in accordance with the classification

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of defects and with Standard MIL-STD-105. Sample discs for destructive test shall be tested as specified in 4.7.2.

4.6.1 Classification of defects.

4.6.1.1 Cover, tank, assembly and detail (Dwg B116-6-90).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing, incorrect, or incorrectly assembled
102	Locating holes incorrectly positioned
103	Machine finish incorrect
104	Welding incorrect
105	Clamping recess incorrect or incorrectly located
106	Hardness incorrect
107	Damage (see 3.7)
Minor:	AQL 2.5 percent defective
201	Diameter incorrect
202	Dome incorrectly formed
203	Chamfer missing or incorrect

4.6.1.2 Gasket, adapter (Dwg B116-6-125) and gasket, nozzle (Dwg B116-6-126).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter incorrect
102	Center hole missing or incorrect
*103	Off center hole missing or incorrectly located
104	Bolt holes incorrect or incorrectly spaced
105	Damage (see 3.7)

* Applicable to Drawing B116-6-125 only.

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4.6.1.3 Gage, high pressure line, assembly (Dwg C 116-6-115).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing, incorrect, or incorrectly assembled
102	Tube incorrectly formed
103	Distance between center lines of tube openings incorrect
104	Distance between center of bend and fitting seal incorrect
105	Distance between seals incorrect
106	Nuts frozen on tube
107	Damage (see 3.7)

4.6.1.4 Valve, relief (Dwg C 116-6-119).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Internal thread incorrect
102	External thread incorrect
103	Dimension between center line of internal thread to face of external thread incorrect
104	Dimension between center line of external thread to face of internal thread incorrect
105	Damage (see 3.7)

4.6.1.5 Regulator, pressure (Dwg C 116-6-120).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Pressure reduction incorrect
102	Damage (see 3.7)

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4.6.1.6 Head, safety (Dwg C 116-6-127).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Rupture pressure incorrect
102	Pipe threads incorrect
103	Length of nipples incorrect
104	Overall length incorrect
105	Damage (see 3.7)

4.6.1.7 Adapter, AN 816-12B (Dwg D116-6-106).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Inside diameter incorrect
102	Damage (see 3.7)

4.6.1.8 Preparation for delivery (level A, section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Surfaces not cleaned
102	Preservation missing or incorrect
103	Packaging methods incorrect
104	Containers incorrect
105	Quantity per package or container incorrect
106	Components, repair parts, or technical publications missing
107	Container closure incorrect
108	Container or package damaged (broken, crushed, torn, punctured, or wracked)
109	Strapping loose or missing
110	Cushioning, blocking, bracing missing or insufficient to prevent movement within packages and pack
111	Marking incorrect, missing, or illegible

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4.7 Tests.

4.7.1 Leakage. All tests shall be conducted under standard ambient temperature and pressure conditions.

4.7.1.1 Tank and frame assembly. The tank and frame assembly shall be subjected to a hydrostatic pressure of 100 ± 2 psig for a minimum period of 5 minutes.

4.7.1.2 Assembled disperser, air and agent. A minimum of 29 pounds of powdered technical talc, conforming to Specification MIL-T-50036, shall be placed in the agent hopper and the cover secured in place by means of the hold-down clamps. The pressure container shall be charged with 2000 ± 100 psig of air. The operating valves shall be regulated so that the disperser system is charged and ready for operation. All fittings and joints of the air system (high and low pressure) shall be brushed with a soap solution. Evidence of air bubbles on the air system or talc on the agent system shall constitute leakage.

4.7.2 Safety head rupture pressure. The safety head shall be tested as follows: Connect a source of compressed air or hydraulic pressure adjustable from 0 to 100 psig with a gage in the safety head inlet line. Exhaust the other side of the head to atmospheric pressure. Gradually increase the pressure at a rate of 1 psig per second in the inlet line to 79 psig. The safety head rupture disc must not rupture. Further increase the pressure at the same rate until either the disc ruptures or 100 psig is reached. The safety head rupture disc must rupture before 100 psig is reached.

4.7.3 Functioning. The assembled disperser shall be filled and tested as specified in 4.7.1.2. Performance of the pressure regulator shall be determined by reading the high and low air pressure gages. The assembled disperser shall be function tested by fully opening the ball valve.

4.8 Acceptance/rejection criteria. If any assembled disperser, tank and frame assembly, or pressure container fails to meet the requirements of this specification, that disperser, tank and frame assembly, or pressure container shall be rejected. If any safety head fails to meet the test specified in 4.7.3, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. The disperser components and repair parts thereto shall be cleaned and dried in accordance with the applicable requirements of Specification MIL-P-116, using cleaning method C-1 and any drying method listed therein.

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5.2 Preservation and packaging. Level of preservation and packaging shall be as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Preservation and packaging methods. Unless otherwise specified, preservatives, including the application thereof, and packaging methods cited herein shall be in conformance with the requirements of Specification MIL-P-116.

5.2.1.2 Unit preservation and packaging.

5.2.1.2.1 Gun, portable riot control agent disperser, M9. The gun and components thereto shall be packaged in accordance with the level A requirements of Specification MIL-G-51025.

5.2.1.2.2 Hose assembly. The hose assembly fittings shall be coated with type P-2 preservative compound and wrapped with greaseproof barrier-material conforming to grade C of Specification MIL-B-121, securing the wrap in place with pressure-sensitive tape conforming to type II, class 1 of Specification PPP-T-60.

5.2.1.2.3 Gloves, rubber. Each pair of rubber gloves shall be packaged in either a paperboard folding or set-up box conforming to style II, type A or B, class a of Specification PPP-B-566 or type I, class A, style 3 or 4 of Specification PPP-B-676.

5.2.1.2.4 Straps, cargo. Cargo straps shall be packaged using method III.

5.2.1.2.5 Packing, preformed. Each preformed packing may be coiled a maximum of 3 coils and shall be packaged using method IC-3.

5.2.1.2.6 Wrenches. The wrenches shall be coated with type P-7 preservative and packaged using method IC-1.

5.2.1.2.7 Head, safety. The spare safety head shall be packaged in accordance with method IC-2 of Specification MIL-P-116, using a paperboard folding box, style, type, and class optional, conforming to Specification PPP-B-566 as the container.

5.2.1.2.8 Clamp. All unpainted surfaces of the MS-Type Clamps shall be coated with Type P-7 preservative and packaged in accordance with method IC-2 of Specification MIL-P-116, using a paperboard folding carton, style, type, and class optional, conforming to Specification PPP-B-566 as the container.

5.2.1.2.9 Tie down. All unpainted surfaces of the tie downs shall be coated with Type P-7 preservative and packaged in accordance with method IC-2 of Specification MIL-P-116, using a paperboard folding carton, style, type, and class optional, conforming to Specification PPP-B-566 as the container.

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5.2.1.2.10 Tubing, rubber. The tubing shall be packaged using method III.

5.2.1.2.11 Connector. The spare connector shall be packaged in accordance with method IC-1 or IC-3.

5.2.1.2.12 Technical publications. Technical publications shall be packaged using method IC-1 or IC-3.

5.2.1.2.13 Protective gas mask hoods. The two gas mask hoods shall be packaged together in accordance with method III.

5.2.1.3 Intermediate packaging.

5.2.1.3.1 Items packaged as specified in 5.2.1.2.2 through 5.2.1.2.12. Items, unit preserved and packaged as specified in 5.2.1.2.2 through 5.2.1.2.12, shall be intermediately packaged in a W5c, style optional fiberboard box, having inside dimensions of 10-3/4 by 8-1/2 by 5 inches and conforming to Specification PPP-B-636. Internal cushioning or blocking shall be used as required to prevent movement of items within the container. The box shall be closed by taping in accordance with the appendix to Specification PPP-B-636 for interior boxes.

5.2.1.3.2 Items packaged as specified in 5.2.1.2.1 and 5.2.1.3.1. Items packaged as specified in 5.2.1.2.1 and 5.2.1.3.1 shall be further packaged in a W5c, style optional fiberboard box, having inside dimensions of 10-1/4 by 10-1/4 by 25-3/4 inches and conforming to Specification PPP-B-636. Internal cushioning or blocking shall be used as required to make a tight package. The box shall be closed in accordance with the appendix to Specification PPP-B-636 for interior boxes.

5.2.2 Level C. The disperser and components thereto shall be cleaned and dried as specified in 5.1 and preserved and packaged in accordance with the supplier's commercial practice, with the exception that the M9 disperser gun shall be packaged in accordance with the level C requirements of Specification MIL-G-51025.

5.3 Packing. Level of packing shall be as specified (see 6.1).

5.3.1 Level A. The disperser shall be positioned on the skid base as shown in figure 1. The skid shall be fabricated as shown on figure 2 and the disperser secured to the skid by six steel brackets as shown in figure 3. The brackets shall be secured to the skid by four each 3/8 by 6-1/2 long and eight each 3/8 by 3-1/2 long steel carriage bolts with nuts, plain washers, and lockwashers. The safety head and ball valve outlet openings shall be covered with pressure sensitive tape conforming to type I, class 1 of Specification PPP-T-60. Mounted gages shall be wrapped with cushioning material conforming to type I, class C of Specification PPP-C-843, securing the wrap in place with pressure sensitive tape conforming to type II, class 1 of Specification PPP-T-60. The safety head and ball valve outlets shall be

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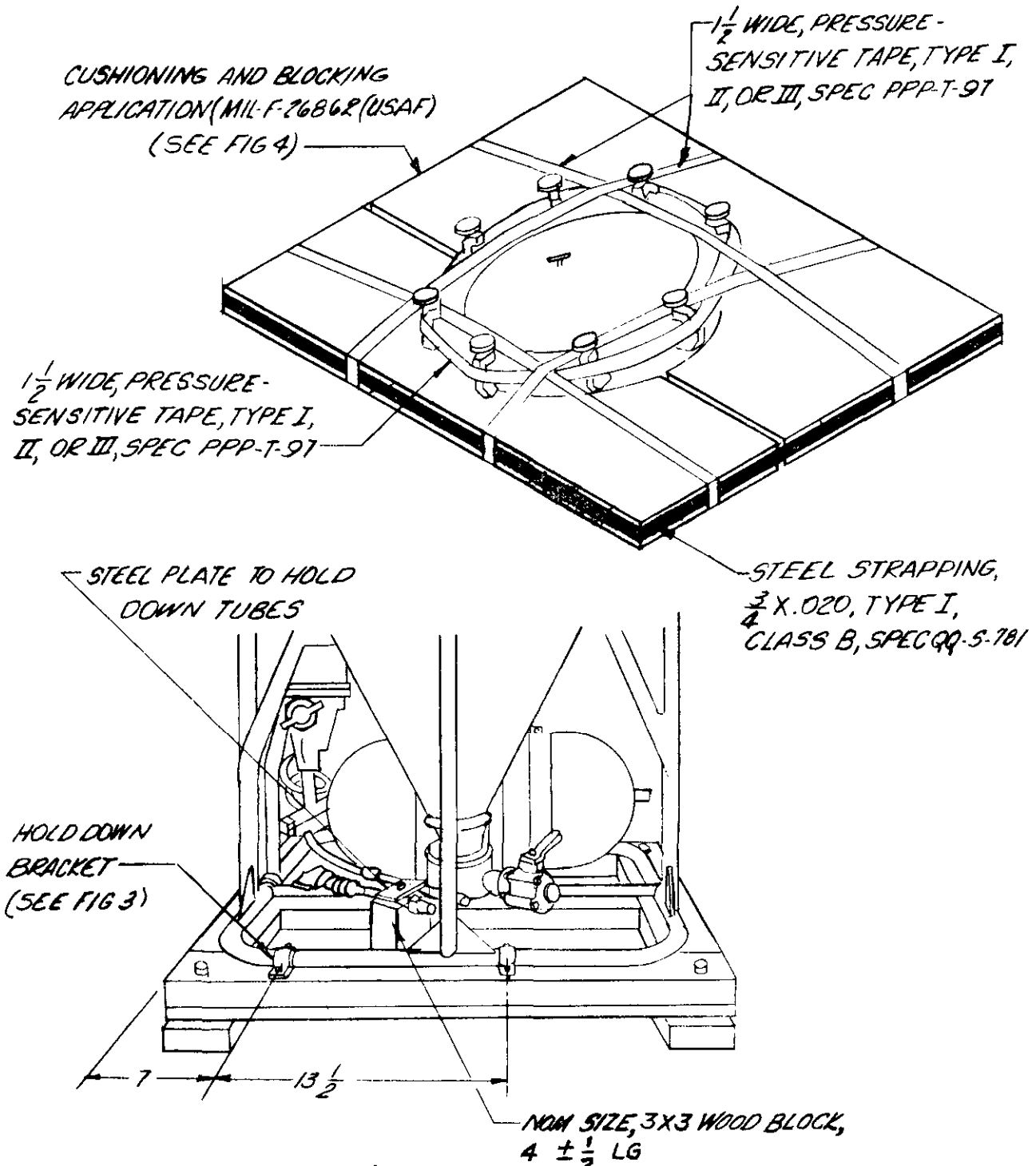
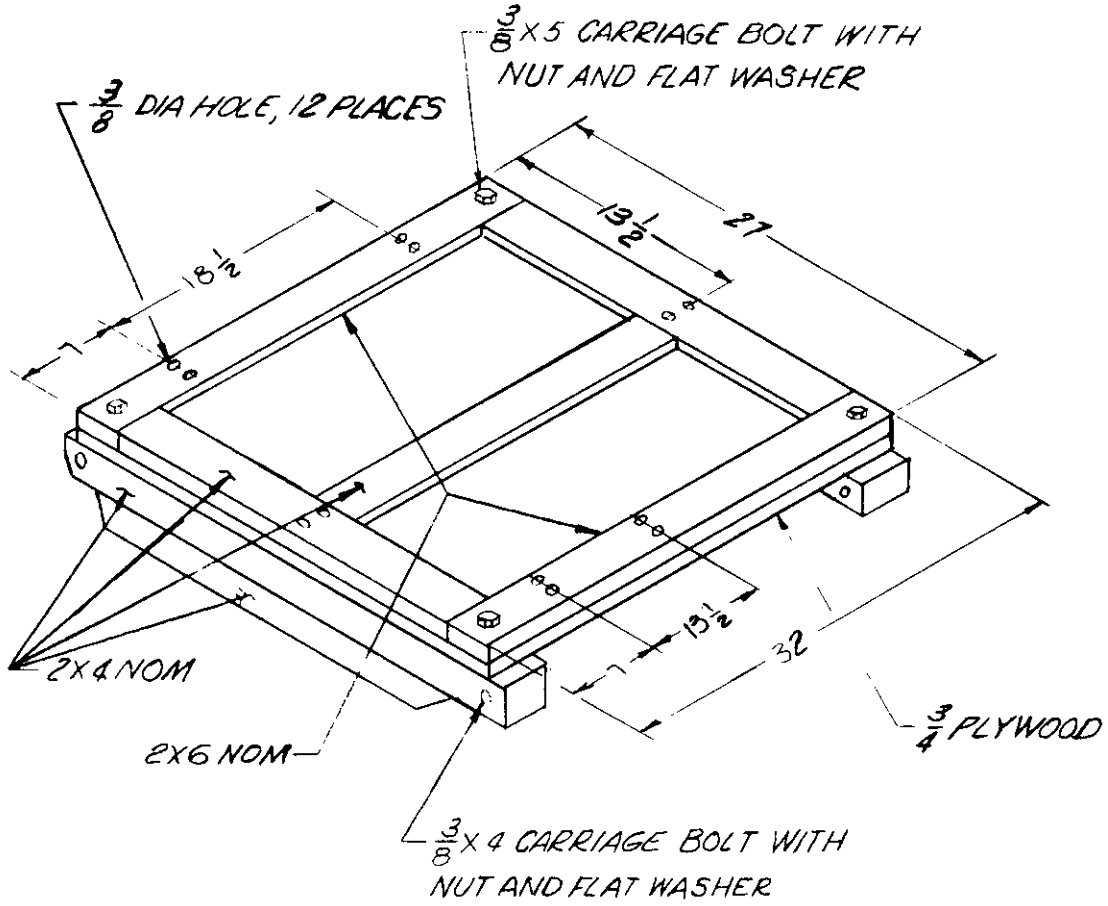


Figure 1

Positioning Of Cushioning, Strapping And Tie-Down Application, Disperser Unit And Hold Down Brackets

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NOTE:

1. USE $\frac{3}{8} \times 3\frac{1}{2}$ BOLT, CARRIAGE WITH NUT AND LOCK WASHER ON 32" SIDE (FF-B-571) AND $\frac{3}{8} \times 4\frac{1}{2}$ BOLT, CARRIAGE WITH NUT AND LOCK WASHER ON 27" SIDE (FF-B-571)

Figure 2

Skid Base

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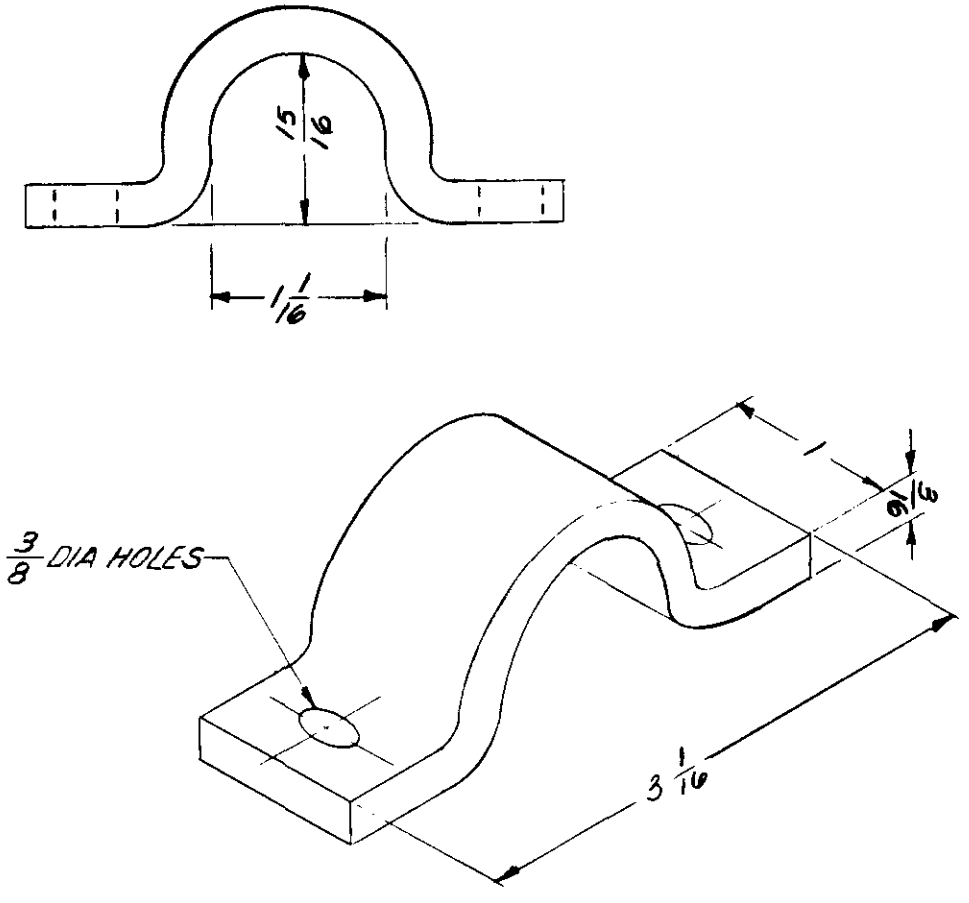


Figure 3
Bracket, Hold Down

Material, Hot Or Cold Rolled Steel, Spec QQ-S-698

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supported and secured in place in the manner shown on figure 1 using a 3 by 3 block 4 inches ± 1/2 inch long; a 1/8 by 1-1/2 by 4-inch steel bar; and a 3/8 by 6-inch size carriage bolt. A circular disc, 14 inches in diameter with a 1-inch hole drilled in the center, shall be fabricated from either a piece of 1/4-inch thick fibrous-felted hard board conforming to type I, class 2 of Specification LLL-H-35 or plywood conforming to type III or IV of Specification NN-P-515. The disc shall be placed at the base on the inside of the hopper. The items, packaged as specified in 5.2.1.3.2, shall be placed inside the hopper on top of the disc. Internal cushioning and blocking shall be provided to prevent movement therein. The hopper cover shall then be secured in place by hand-tightening the cover hold-down clamps. One-half inch wide filament reinforced tape, conforming to type II of Specification PPP-T-97, shall be wrapped around the outside of the cover hold-down clamps as shown on figure 1. Fiberboard cushioning material, fabricated as shown on figure 4, shall be positioned around the disperser and secured in place in the manner shown in figure 1, using the filament reinforced tape. The wood-cleated plywood cover, fabricated as shown on figure 5, shall be placed over the disperser and secured to the skid base with lag bolts.

5.3.2 Level C. The disperser and components thereto, packaged as specified in 5.2, shall be packed in such a manner as to insure carrier acceptance and safe delivery at first destination for immediate use. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.4 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

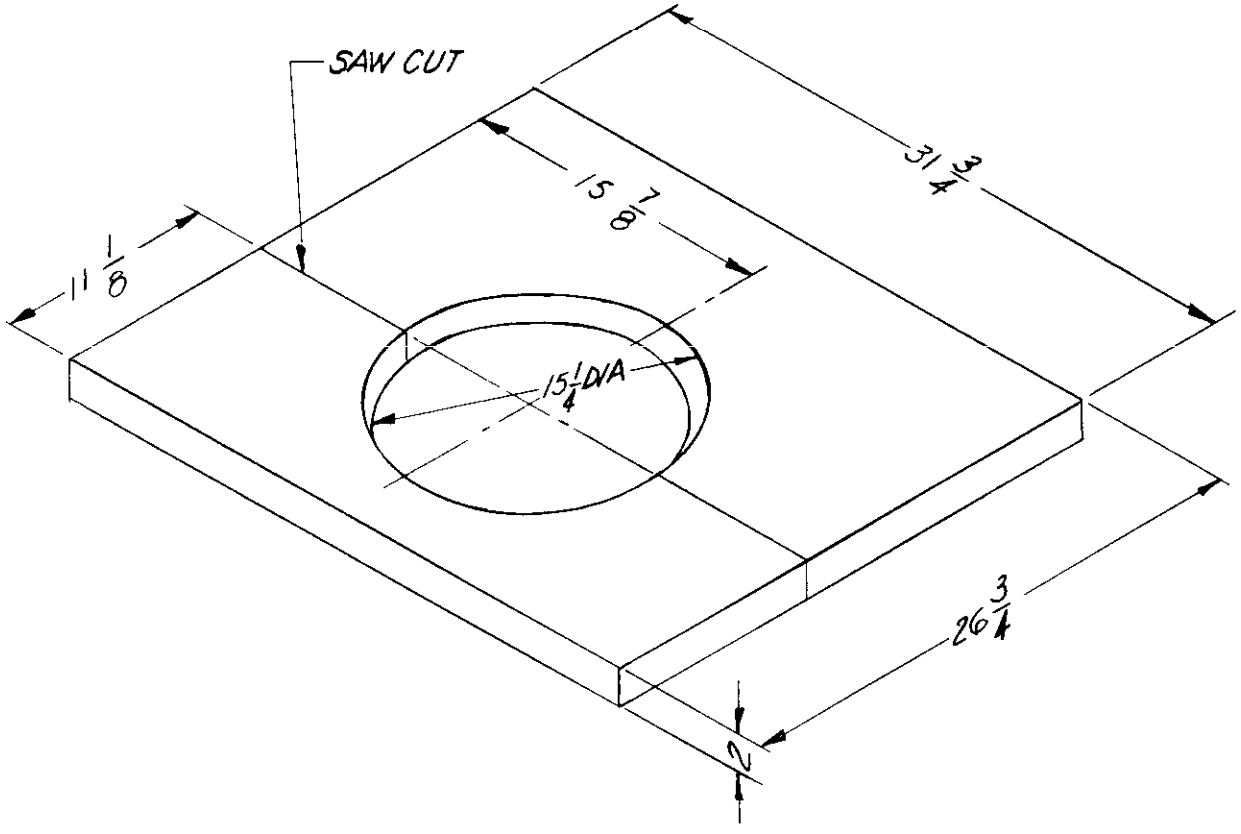
(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.

6.2 Component specification. The following specification is required for the manufacture of the disperser:

MIL-G-51025 - Gun, Portable Riot Control Agent Disperser, M9.

6.3 Supersession data. This specification supersedes Chemical Corps Purchase Description 196-31-105, dated 31 May 1960.

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Material, Fiberboard, Solid Dunnage, Multipurpose Cushioning And Blocking Application, Type II, Spec Mil-F-26862

Figure 4

Celotex Pad For Blocking And Bracing

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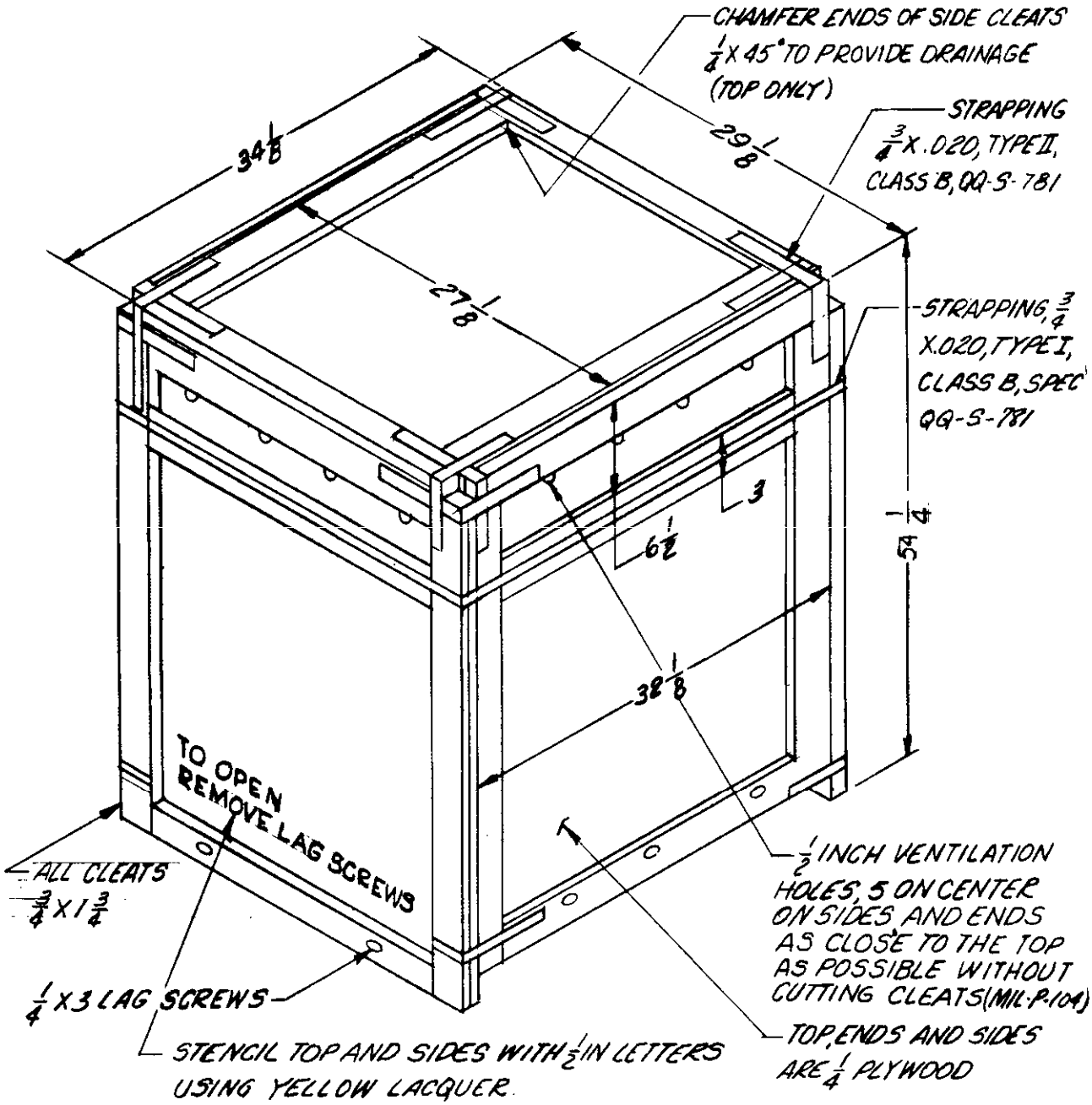


Figure 5

Modified Style B Box Wood Cleated-Plywood (PPP-B-601 Using Skid Base)

TOLERANCE ± 1/8

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Custodian:

Army - Chemical Corps

Preparing activity:

Army - Chemical Corps

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